MAINTAIN BASIC STORAGE

Storage concepts

- **Partition**
  Disk divided into smaller parts
- For taking backup
- For improving performance

PARTITION DISKS ON REDHAT SYSTEMS

STEP1

- Identify the firmware of the system [BIOS/UEFI]
- To identify the firmware type, we can use the #dmidecode

STEP2

- Suppose Bios is ours then we using method is Master Boot Record(MBR)
- UEFI time we using GUID Partition Table(GPT)

STEP3

- Partitioning using appropriate tools.
- In MBR case we use tools like #fdisk to create and manage partitions within the MBR scheme.
- For GPT partitioning, we use tools like #gdisk, which is similar to fdisk but designed specifically for GPT partitions.
- **MBR (Master Boot Record)**
- Used with BIOS firmware.
- Supports 4 primary partitions OR 3 primary + 1 extended partition.
- Maximum disk size: approximately 2 TB.
- **GPT (GUID Partition Table)**
- Commonly used with UEFI firmware.
- Supports up to 128 partitions on Windows by default.
- Supports disks larger than 2 TB.
- **STEP 4**
- Types of partitions
- **Normal partition:-** This is a standard partition used for storing data or installing an operating system.
- **Swap partition**:- its related to system Ram.This type of partition is used as virtual memory,allowing the system to move data between RAM and disk storage when physical memory (RAM) is full.
- **LVM(Logical Volume Management):-** in industry purpose we commonly using LVM and it allows for flexible management of disk space by creating volumes that can span multiple physical disk.
- **STEP5**
- Initialize kernel

- - after creating partitons,may need to initialize the kernel.
- #partprobe
- **STEP6**
- Format drives
- Ext2,ext3,ext4 and XFS
- #mkfs.ext4 /dev/sdb1
- **STEP7**
- After we want mount for use that space
- Tem mount or permanent mount
- Mount /dev/sdb1 /data (this is tem mount)
- df -Th
- umount /data
- #lsblk
- Vim /etc/fstab(for permanent mounting)
- /dev/sdb1 /data ext4 defaults 0 0
- Systemctl daemon reload (for activate this mount)
- mount -a (for activate this)
- df -Th
- for delete a disk
- #fdisk /dev/sdb
- d
- w
- partprobe

SWAP PARTITION:-

- example:- imagine our system using 2 gb ram and is fully loaded and our storage is free we allocate that space to ram 2 gb we can convert to 4 gb ram that is called swap partition.
- This is a temperrory solution.
- Virtual memory = RAM+SWAP

RAM Size	Swap Space Size
2 GB	2 GB
4 GB	4 GB
8 GB	4 GB
16 GB	8 GB

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- free -mt to check Ram space
- in swap after partition inform to kernel and mkswap /dev/sdb1
- /dev/sdb1 swap swap defaults 0 0

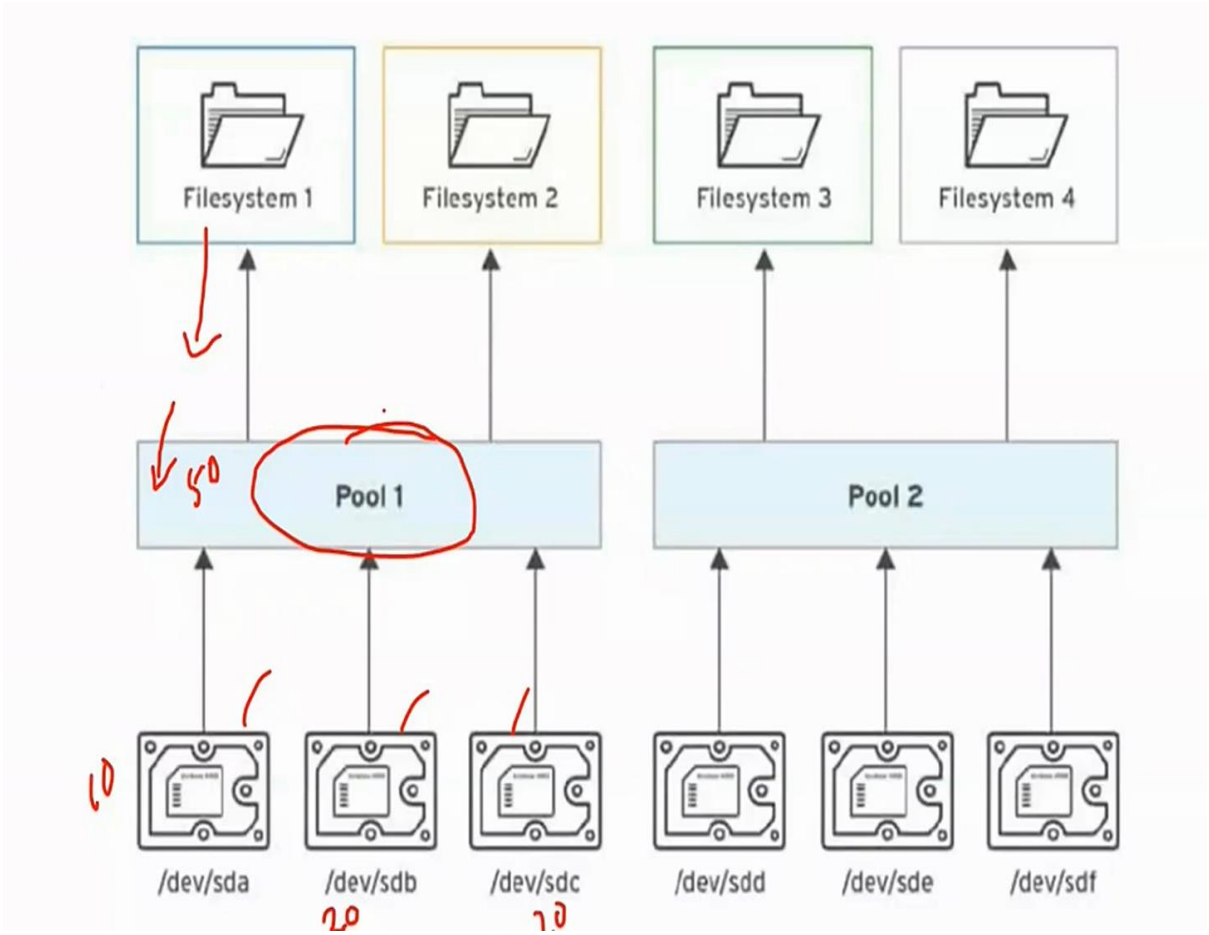
- For activate swapon -a this will activate all or swapon /dev/sdb1
- For off swapoff
- free -mt
- note:-
- when install epel
- in aws a issue comes like killed
- the issue means Ram doesn't have space
- for partition we can use fdisk or parted to partiion both can work in gpt and mbr very well.
- df -hT shows mounted filesystems, their types, and available space.

LVM (MANAGE LOGICAL VOLUME)

- without data lost we can extend it.
- 1. Start fdisk
- After partition and type
- We want create a physical volume
- Pvcreate /dev/sdb1
- Maybe an error occur like command not found means package not installed
- Yum install -y lvm2
- Pvdisplay
- Next we want to create volume group
- Vgcreate myvg /dev/sdb1
- Vgdisplay
- And third step is logical volume
- Lvcreate -L 600 -n mylv myvg
- Lvdisplay
- After this normaly mount
- Mkfs.ext4 /dev/myvg/mylv
- Mkdir /data
- Systemctl daemon-reload
- Mount -a
- For extend
- Lvextend -L 700M /dev/myvg/mylv
- (ext4) resize2fs /dev/myvg/mylv
- (xfs) xfs_growfs /data
- Lvreduce -L 500M /dev/myvg/mylv

STRATIS

- Simplify storage
- This is third party service



- This is the working of stratis
- Installation
- #yum install -y stratisd stratis-cli
- For this we need a dedicated drive
- Startis pool create pool1 /dev/sdb
- Stratis pool list
- To add another space to this pool
- Stratis pool add-data pool1 /dev/sdc
- Startis filesystem create pool1 fs1
- Blkid (for get this uuid)
- Blkid /dev/stratis/pool/fs1
- After mount this
- Same we did previously
- Vim /etc/fstab

```
UUID=d0be04e8-fc37-4928-af70-fc3d6595ef71       /       xfs     defaults        0
  0
UUID=a2e9d044-9192-497f-a8c6-27aa7940de18       /boot   xfs     defaults        0
  0
UUID=7B77-95E7  /boot/efi       vfat    defaults,uid=0,gid=0,umask=077,shortname=winnt
  0       2
UUID=d3801d6f-f6a5-4b69-a6b2-86c5e887ac7e /mnt/stratis  xfs  defaults,x-systemd.requir
es=stratisd.service 0   0
```

- Systemctl daemon-reload

- Mount -a
- df -Th or lsblk
- also we can take a snap short
- #stratis filesystem snapshot pool1 fs1 snap1
- #Stratis filesystem list (after we can see both snap and og)
- Umount /data -> remove from fstab->systemctl daemon-reload ->startics filesystem destroy pool1 fs1
- Stratis filesystem destroy pool1 snap1
- Lsblk
- Stratics pool destroy pool1
- lsblk

LVM (Logical Volume Manager): LVM is a Linux storage management system that allows us to create, extend, and reduce logical volumes easily.

- **PV (Physical Volume): The disk or partition prepared for LVM.**
- **VG (Volume Group): A pool of storage created from one or more PVs.**
- **LV (Logical Volume): A virtual partition created from a VG**.

STRATIS: Stratis is a Linux storage management solution that simplifies storage administration by providing features like pooling, snapshots, and dynamic storage management.